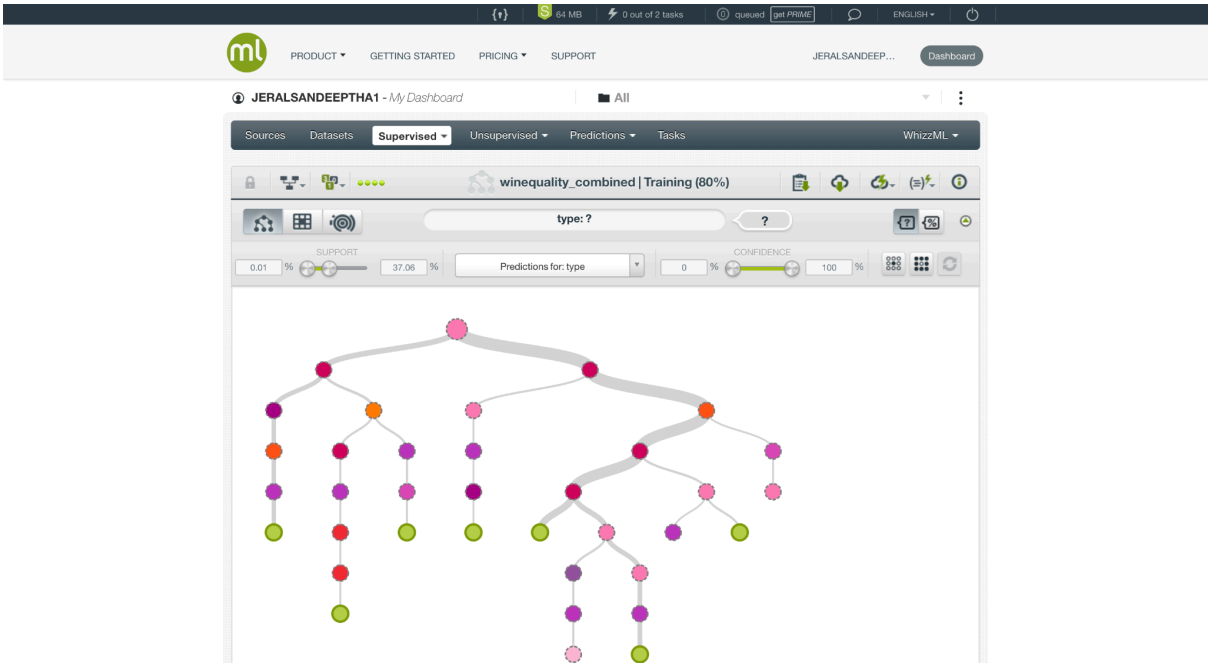
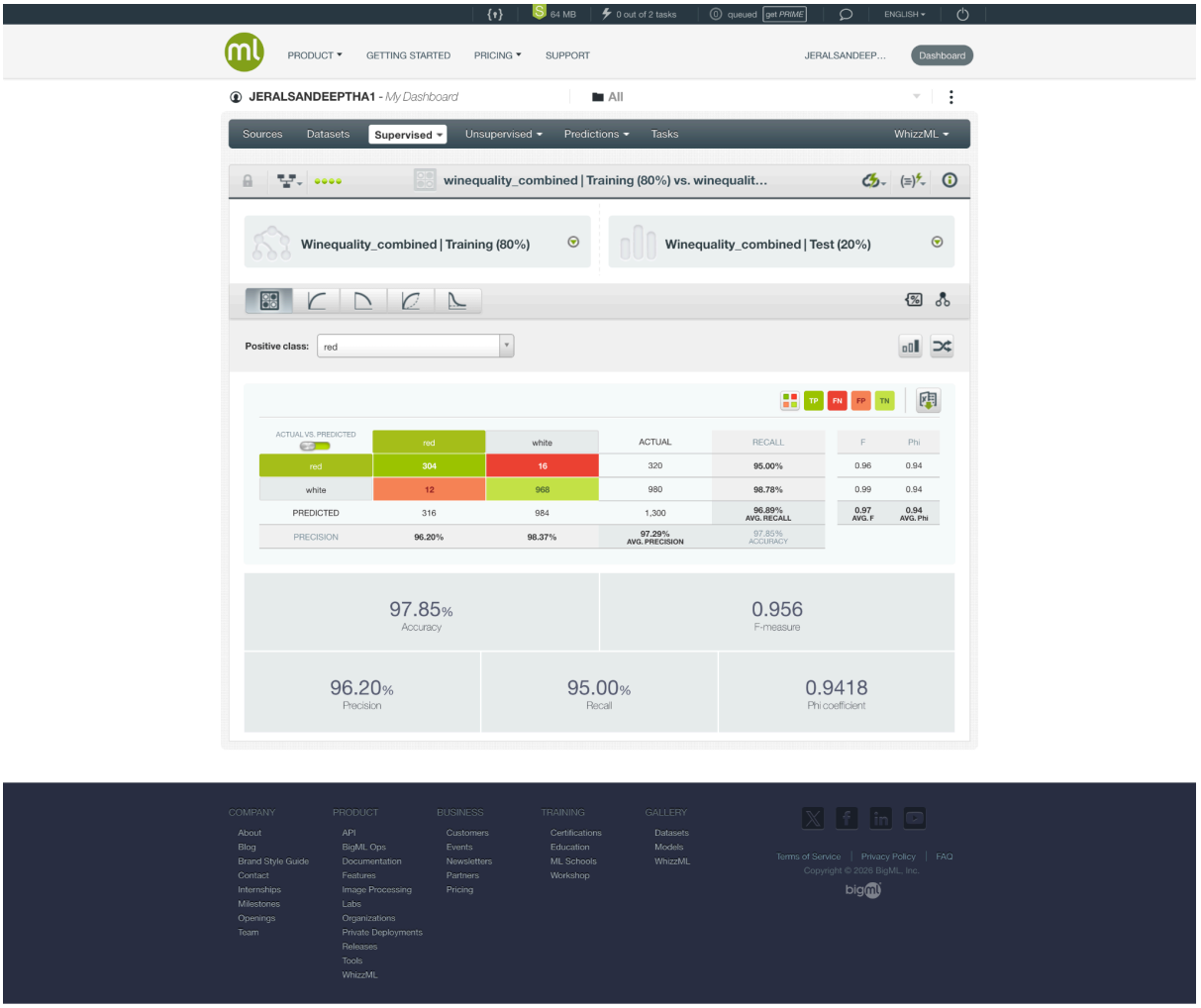
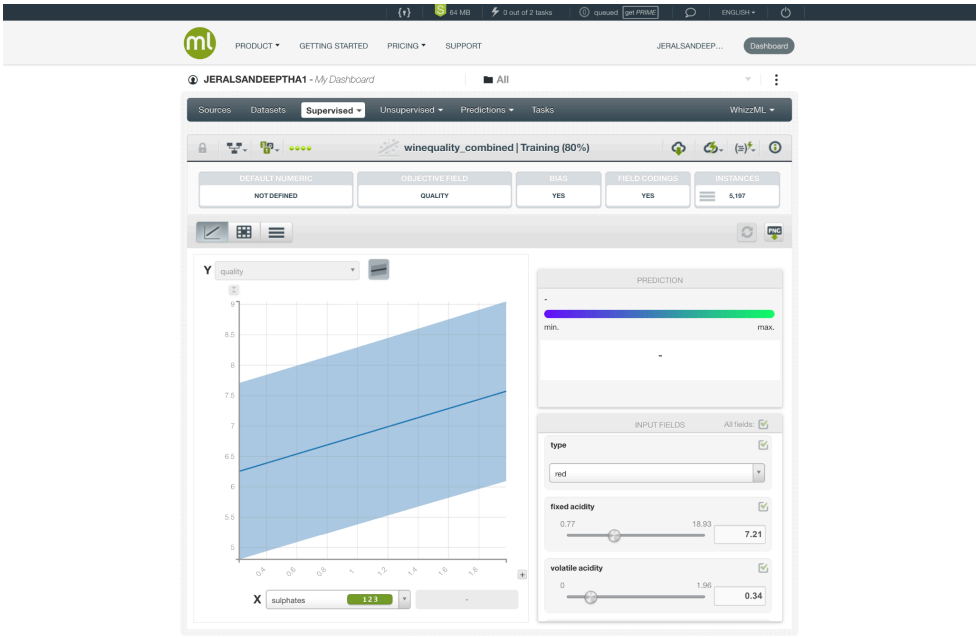
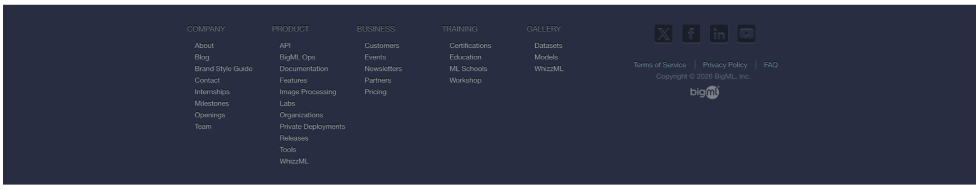
Task A: Classification

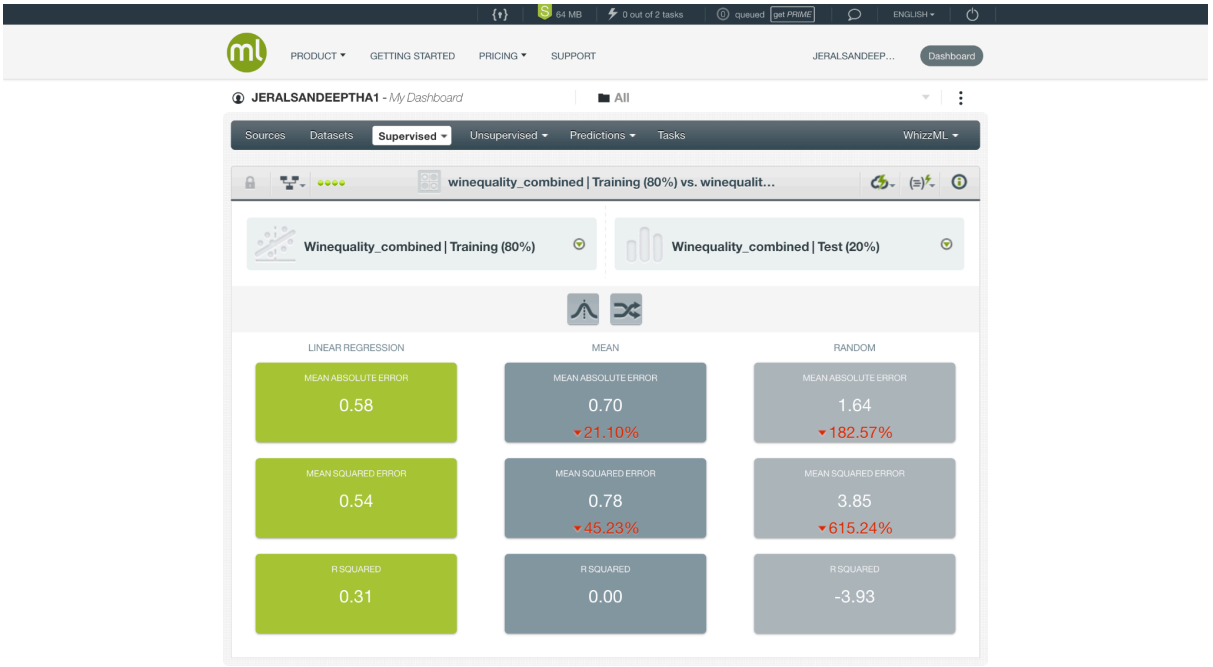


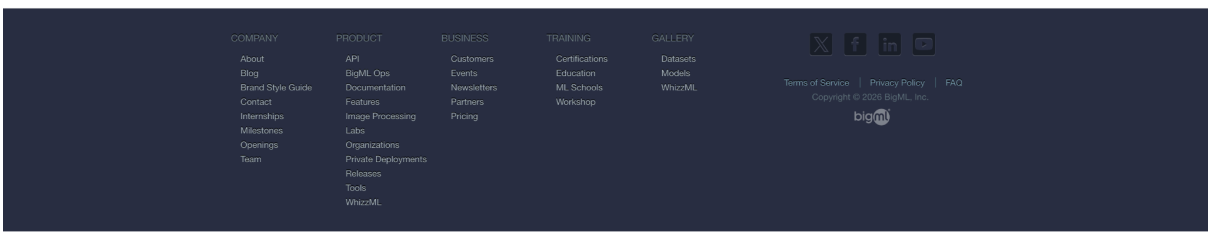


Task B: Regression









Structured Questions

1. What are the input features and the target variable in the classification task?

The input features are physicochemical measurements of the wine (such as acidity, sugar, pH, alcohol, etc.). The target variable is type (which **wine_type**).

2. Why is predicting wine_type a classification problem?

Because the output variable consists of discrete, categorical classes. In here red and white.

3. State the accuracy obtained by your classification model. What does accuracy represent?

Accuracy: 97.85%

It represents the proportion of total predictions that the model got right. It measures how often the model correctly classified a wine sample which is both red and white combined, out of all the total test instances evaluated.

4. According to the confusion matrix, which wine type had more correct predictions?

White wine.

We can say that by looking at the true positive diagonals of the confusion matrix. Basically, in white wine, it has 968 correctly predicted samples and on the other hand in red wine, it has 304 correctly predicted samples.

5. Why is predicting quality a regression problem?

Because the target variable is a continuous numerical score rather than a categorical class label. Basically we have numerical numbers for target variable which we can't categorize into a specific label.

6. State the MAE obtained by your regression model. What does a lower MAE indicate?

MAE Obtained value is 0.58 (as shown under Linear Regression).

It measures the average size of the prediction errors without considering their direction. A lower MAE indicates higher prediction accuracy and its meaning is that the model's predictions are, on average, closer to the actual true values in the test dataset.

7. Why is the dataset divided into separate training and testing datasets?

So the model can train on one subset and be independently evaluated on unseen testing data to measure its generalizability and real world performance.